Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

SKILLPILOT AI

LLM POWERED CAREER COACH

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PROJECT GUIDE

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INTRODUCTION

In today's fast-evolving job market, students and professionals face significant challenges in choosing the right career path and upskilling themselves effectively. With the rise of automation, AI, and rapidly changing industry demands, many individuals struggle to find personalized career guidance. Traditional career counseling services are often limited in accessibility, time-consuming, and not tailored to individual strengths or aspirations. Online resources exist, but they are usually fragmented, generic, or outdated, leaving learners confused about what skills to acquire and how to align them with their career goals.

During real-time observation, it is evident that many students preparing for jobs feel lost while navigating career decisions, especially when balancing between academic choices and industry expectations. Similarly, working professionals aiming for career transitions lack structured guidance and spend hours searching for relevant resources without clarity. These gaps strongly motivated us to explore how AI, particularly Large Language Models (LLMs), can be leveraged to provide personalized, scalable, and accessible career coaching for individuals at different stages of their professional journey.

LITERATURE SURVEY OF EXISTING SYSTEMS

Title (Year)	Authors	Key Contribution
Personalized Career Recommendation System using LLMs (2024)	Li, Zhang & Kumar	Developed an AI system leveraging large language models to analyze resumes and recommend personalized career paths; enhances decision-making for students and professionals.
AI-Based Career Guidance Chatbot for Students (2023)	Sharma et al.	Created a chatbot that interacts with students to assess skills and suggest suitable careers; integrates NLP techniques to provide real-time guidance.
SkillBridge: AI-Driven Career Transition Platform (2022)	Chen, Singh & Patel	Designed a platform that identifies skill gaps and suggests learning paths for career transitions; uses machine learning to match job market demands.

LITERATURE SURVEY OF EXISTING SYSTEMS

Title (Year)	Authors	Key Contribution
Career Compass: AI-Powered Soft Skills Profiling (2021)	Johnson & Lee	Developed a system incorporating soft skills assessment into career recommendations; improves personalization and holistic career guidance.
AI-Powered Career Path Prediction for CS Students (2020)	Kumar et al.	Proposed a deep learning model predicting likely career paths based on academic and skill profiles; tailored for computer science and software engineering students.
Intelligent Career Advisor System (2020)	Nguyen & Brown	Implemented an AI system providing personalized career guidance using decision trees and NLP analysis of student profiles; supports long-term career planning.

LIMITATIONS OF EXISTING SYSTEMS

Title (Year)	Authors	Abstract	Limitations
LABOR-LLM: Language-Based Occupational Representations with LLMs (2024)	Susan Athey, Tim Fu, Xinzhu Li, Anson Sui	Proposed LABOR-LLM, a language-based occupational representation fine-tuned with resume-like data to predict next jobs. Showed superior performance compared to traditional foundation models.	Pre-training data not fully representative; may face generalizability issues across diverse labor markets.
A Systematic Literature Review of AI in Coaching (2024)	Silvia P. D'Ambrosio, Rita Bissola, Guido Capaldo, Susanna N. L. Piccareta	Reviewed peer-reviewed studies on AI in coaching. Found that LLMs can enhance accessibility, affordability, and complement human coaches.	Most studies are early-stage; LLMs exhibit problematic behavior; lack of longitudinal effectiveness research.
Leveraging LLMs in Professional Coaching (2024)	Henning Brütting	Investigated fine-tuning LLMs with coaching frameworks (e.g., PEARLS), activity-text prompts, and enterprise solutions like MobileCoach.	LLMs lack empathy, sensitivity to non-verbal cues, and robust evaluation; limited scalability/generalizability.

LIMITATIONS OF EXISTING SYSTEMS

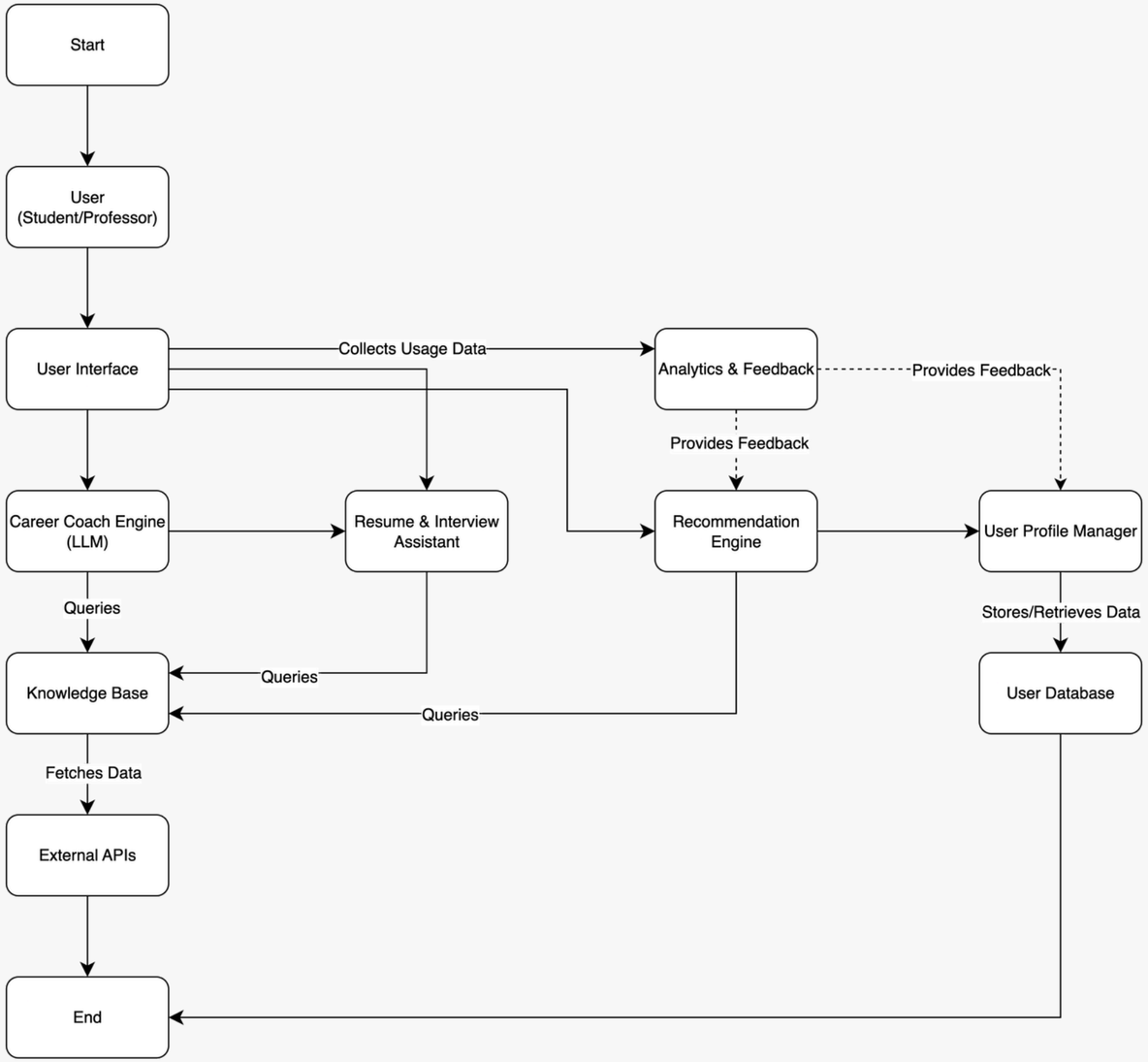
Title (Year)	Authors	Abstract	Limitations
Artificial Intelligence in Career Counseling: A Test Case with ResumAI (2023)	Muhammad Rahman, Md. Rahat Hossain	Introduced ResumAI, an AI-based resume feedback tool. Conducted literature review on ethical AI and tested functionality through a prototype web app.	Limited user evaluation; shallow exploration of ethics; scalability not studied.
Fair Representation Learning for Heterogeneous Information Networks (2021)	Ziqian Zeng, Yushun Dong, Jiebo Luo, Fenglong Ma	Developed methods (sampling-based, projection, GNNs) to de-bias heterogeneous network representations with application in automated career counseling.	Focused narrowly on algorithmic fairness; real-world deployment and broader systemic issues not covered.
User Acceptance of Gender Stereotypes in Automated Career Recommendations (2021)	Clarice Wang, Ana-Andreea Stoica, Celia Chen, Augustin Chaintreau	Designed a fair recommender system using gender-debiasing techniques. Despite maintaining accuracy, users preferred biased recommendations.	Human biases persisted, leading to rejection of fairer models; limits adoption in real-world scenarios.

PROBLEM STATEMENT & OBJECTIVE

Choosing the right career path has become increasingly difficult in today's dynamic job market. Students and professionals face challenges in identifying suitable opportunities, mapping required skills, and accessing reliable guidance. Existing solutions such as traditional counseling services and online career portals are either too generic, fragmented, or inaccessible to many due to high costs and limited personalization. AI-powered career tools exist, but they often lack adaptability, real-time interactivity, and contextual understanding of individual aspirations.

This project aims to address these challenges by developing an LLM-powered career coach that provides personalized, adaptive, and accessible career guidance. The solution will focus on bridging the gap between user needs and current offerings by offering tailored career recommendations, skill development pathways, and practical support for resume building, interview preparation, and continuous upskilling.

SYSTEM DESIGN



IMPLEMENTATION

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THANK YOU